Gene panel or exome? A published Indian case: diarrhoea, thick skin and stiff joints that no single panel would have covered

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Nayek S, Sandhu A, Kumar A, Malik S. *Infantile Systemic Hyalinosis: A Familiar Symptom Unveiling an Unusual Disease.* Cureus 2026;18(2):e103465. doi:10.7759/cureus.103465. Department of Pediatrics, All India Institute of Medical Sciences (AIIMS), Bhopal. Open access (CC BY 4.0). doi.org/10.7759/cureus.103465

The patient

- Girl, 7 months; third-degree consanguineous parents
- 2 months of watery diarrhoea, failure to thrive, severe acute malnutrition
- Diffusely thickened skin; hyperpigmentation over the metacarpophalangeal joints
- Bilateral elbow contractures; markedly reduced movement at shoulders, wrists, hips and ankles
- Gingival hypertrophy; nodular lesion and thickening of the ear
- Anaemia (Hb 6.3 g/dL), leucocytosis, raised CRP, hypoalbuminaemia

Work-up and result

- Stool examination and ova/parasites: no significant finding
- Abdominal ultrasound normal; intestinal biopsy: no villous atrophy, dysplasia or dilated lymphatics
- Stool electrolytes normal (against cystic fibrosis and congenital chloride diarrhoea)
- Skin biopsy declined by the parents
- Whole exome sequencing: **homozygous pathogenic frameshift variant in ANTXR2** (c.1074del)
- Diagnosis: **infantile systemic hyalinosis** (autosomal recessive)

Why exome fitted, and what followed

The features spanned the gut, skin, joints, gums and ears, and the common causes of chronic diarrhoea had been excluded. A picture this broad does not map to one obvious gene panel, which is where exome sequencing helps. After the diagnosis, care was supportive: adequate-calorie nutrition, **albumin infusions** and **physiotherapy**. Diarrhoea was controlled and weight stabilised, but motor improvement was minimal; the authors note there is no permanent cure. The parents were **counselled on the disease, its inheritance and the recurrence risk**. The authors highlight how a rare disease presenting with common symptoms leads to diagnostic delay, and the definitive role exome sequencing played.

Talking points for the doctor visit

- Clear phenotype with a known gene list: a panel is usually enough and gives fewer uncertain results.
- Broad, multi-system or unclear phenotype: consider exome, ideally as a trio.
- Counsel before testing on VUS and secondary findings; take written consent.
- Prenatal testing in India: written consent under the PCPNDT Act; disease testing only; fetal sex never disclosed.

For healthcare professionals' general awareness. Not a substitute for clinical judgement; facts and guidelines may change. Case details are as published by the original authors; consult the full article.

EPISODE 8 · QUICK GUIDE

Gene panel or exome: how to choose

Topic	Key points
What they are	Panel: selected genes for one clinical question (tens to a few hundred). Exome: the coding parts of about 20,000 genes, about 1-2% of the genome, where most known disease-causing variants lie.
Choose a panel	Clear clinical picture with a known set of genes (e.g. hearing loss, cardiomyopathy, hereditary cancer risk). Usually deeper coverage of those genes, fewer VUS and fewer unrelated findings.
Choose exome	Broad or non-specific picture: developmental delay/intellectual disability, multiple congenital anomalies, multi-system disease. ACMG (2021) suggests exome or genome as a first- or second-tier test for congenital anomalies, developmental delay or intellectual disability.
Virtual panel and reanalysis	Sequence the exome but analyse a gene list first; widen the analysis if negative. Exome data can be reanalysed later as new genes are described.
Trio	Sequencing the parents with the child identifies de novo changes faster and gives a clearer report (see Episode 3).
Indian data	78 children with neurodevelopmental disorders, clinical exome (SCTIMST, Thiruvananthapuram): pathogenic/likely pathogenic 41%; VUS 33.3% (Valaparambil et al. 2025).
Trade-offs	More genes, more VUS. Secondary findings (important results unrelated to the question) are possible; counsel first and record consent on what to report.
What both can miss	Large deletions/duplications (use CMA), repeat expansions such as Fragile X, deep intronic changes, methylation (imprinting) disorders, low-level mosaicism. A negative exome does not rule out a genetic disease.
Sample	EDTA blood (parents too for a trio). Prenatal: amniotic fluid or CVS.

References

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